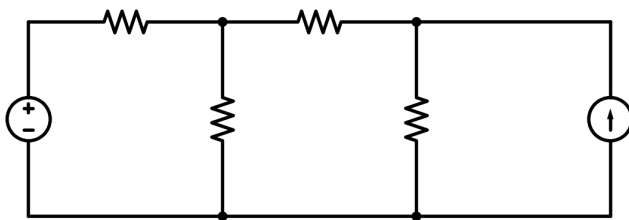


Practice sheet: Node Voltages and Mesh Currents

This sheet is not marked for correctness, and AI tools are allowed. The Part 1 device-free quiz is built from twins of these questions, so make sure you can do each one on your own.

- **Two-tier items** ask for an answer *and* a reason. Both must be right.
- **Predict first:** where asked, write your prediction before you calculate.
- **Confidence:** next to each answer, write S (sure) or U (unsure). This is not marked; it shows you what to revise.
- **Conventions:** node voltages are measured from the reference node, marked with the ground symbol. In KCL, count currents leaving a node as positive. Mesh currents run clockwise; in KVL, count voltage drops as positive. Check every answer: KCL must hold at every node, and the powers must add to zero.

S1. How many equations? (two-tier)



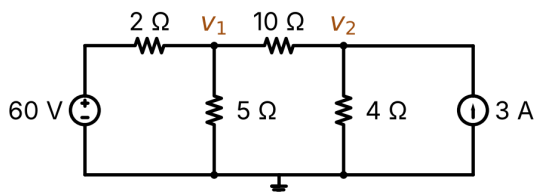
How many equations does each method need?

- Node voltages 3, mesh currents 3
- Node voltages 3, mesh currents 2
- Node voltages 2, mesh currents 2
- Node voltages 2, mesh currents 3

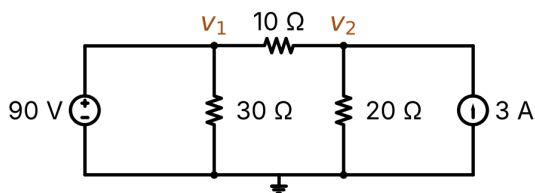
Reason:

- One KCL equation for every essential node, the reference included.
- Every mesh needs its own KVL equation, whatever it contains.
- KCL at every essential node but the reference; a current source in one mesh fixes that mesh current.
- Count every node, including those where only two elements meet.

S2. The node-voltage method.



- Write the KCL equations at nodes 1 and 2.
- Solve for v_1 and v_2 .
- Find the power of each source, say whether it absorbs or delivers, and check the power balance.

S3. A source to the reference (two-tier).

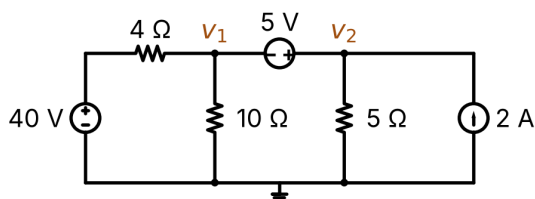
With the bottom node as the reference, how many node-voltage equations must you solve?

- a) 2 b) 1
c) 3 d) 0

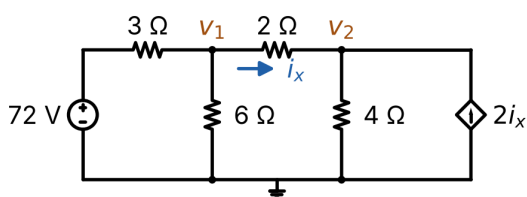
Reason:

- i) Voltage sources are never part of a node equation.
- ii) The 90 V source fixes $v_1 = 90$ V, so only v_2 is unknown.
- iii) Every essential node needs its own equation.
- iv) The 3 A source fixes v_2 .

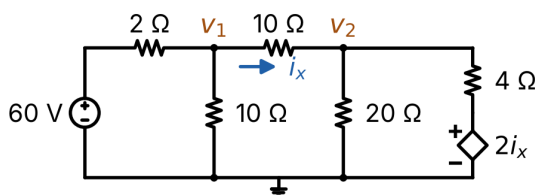
Follow-up: find v_2 and the current the 90 V source delivers. Why does the 30 Ω resistor not appear in your equation for v_2 ?

S4. A supernode.

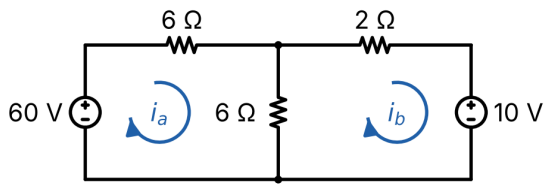
- a) Draw the supernode. Write its KCL equation and its constraint equation.
- b) Find v_1 and v_2 .
- c) Find the power of the 5 V source and say whether it absorbs or delivers.

S5. A node equation with a dependent source.

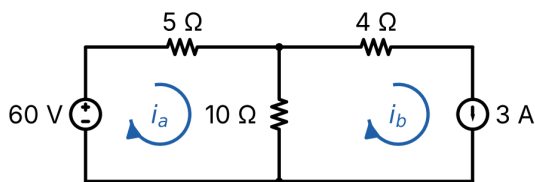
- a) Write the two KCL equations and the constraint equation for i_x .
- b) Find v_1 , v_2 and i_x .
- c) Find the power of the dependent source and say whether it absorbs or delivers.

S6. A dependent voltage source. As in Nilsson Example 4.4.

- a) Write the node equations and the constraint equation.
- b) Find v_1 , v_2 and i_x .
- c) Find the current in the 4 Ω resistor and the power of the dependent source.

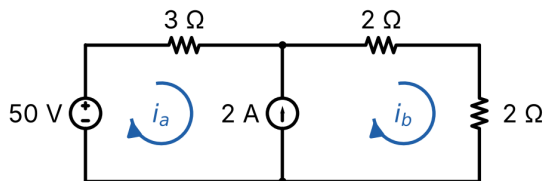
S7. The mesh-current method.

- Write the two mesh equations.
- Find i_a and i_b , and the current in the $6\ \Omega$ middle resistor.
- Find the power of each source and say whether it absorbs or delivers.

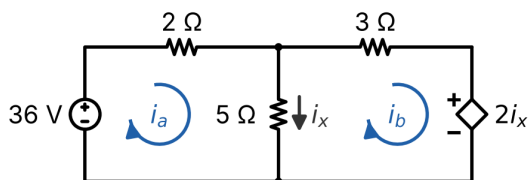
S8. A current source in one mesh (predict first).

Predict first: how many mesh equations do you need?

- Find i_a and i_b .
- Find the voltage across the 3 A source (+ at its top). Does it absorb or deliver power?

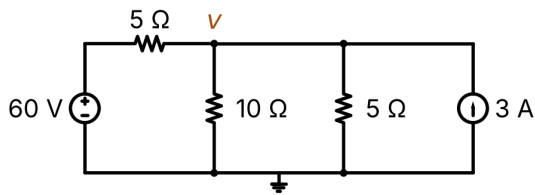
S9. A supermesh.

- Write the supermesh KVL equation and the constraint equation.
- Find i_a and i_b .
- Find the voltage across the 2 A source and its power.

S10. A mesh equation with a dependent source.

- Write the two mesh equations and the constraint equation for i_x .
- Find i_a , i_b and i_x .
- Find the power of the dependent source and say whether it absorbs or delivers.

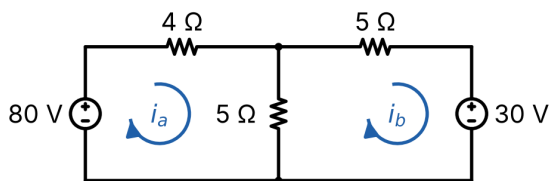
S11. Spot the error. The worked solution below finds v and the power of the 60 V source. It contains the kind of direction slip AI chat tools often make.



1. Reference: bottom node; one unknown, v .
2. KCL at the top node, currents leaving:
 $(v - 60)/5 + v/10 + v/5 - 3 = 0$, so $v = 30$ V.
3. The current in the 5 Ω resistor from the node toward the source is $(30 - 60)/5 = -6$ A.
4. So 6 A flows into the source's + terminal, and the 60 V source absorbs 360 W.
5. Check: $v = 30$ V is less than 60 V, so the answer is reasonable.

- a) Which is the first wrong line?
- b) Correct it.
- c) Check the corrected answer with a power balance.
- d) Line 5 calls itself a check. Why does it prove nothing?

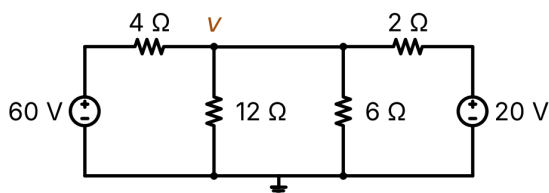
S12. Spot the error in a mesh solution.



1. Mesh a, clockwise, drops positive:
 $-80 + 4i_a + 5(i_a - i_b) = 0$.
2. Mesh b, clockwise, drops positive:
 $5i_b + 30 + 5(i_a - i_b) = 0$.
3. Line 2 simplifies to $5i_a = -30$, so $i_a = -6$ A, and line 1 then gives $i_b = -26.8$ A.
4. Both currents are negative, so both meshes really circulate counterclockwise.

- a) Which is the first wrong line?
- b) Correct it and find i_a and i_b .
- c) Check your answer with a power balance.

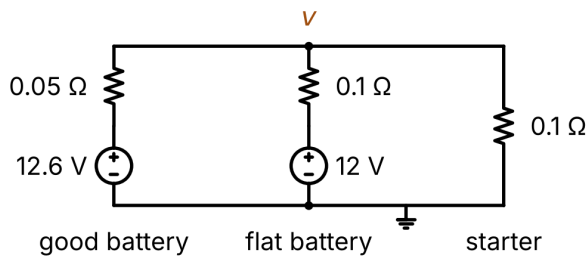
S13. Node voltages or mesh currents?



- a) How many equations does the node-voltage method need? How many does the mesh-current method need?
- b) Solve with the method that needs fewer, and find v .
- c) Find the current in the 20 V source. Is the source being charged?

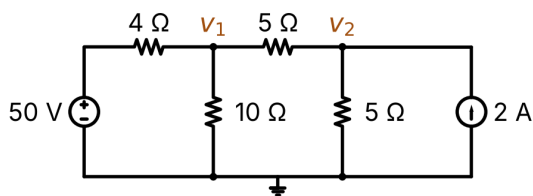
S14. Work backwards. In the S7 circuit, replace the 10 V source with an unknown source V (+ at top). What value of V makes the current in the 2 Ω resistor zero?

- | | |
|---------|---------|
| a) 30 V | b) 60 V |
| c) 10 V | d) 15 V |

S15. Jump-starting a car.

Each battery's resistance includes its share of the jumper cables.

- Before cranking, the starter is not connected. Write the node equation and find v . What current flows in the flat battery, and in which direction? Is it being charged?
- Predict first:* while the starter cranks, does the flat battery help or keep charging? Then find v , the starter current and the current in each battery.

S16. Check a solution without solving again.

Two classmates report their answers. Student 1: $v_1 = 30$ V, $v_2 = 20$ V. Student 2: $v_1 = 28$ V, $v_2 = 24$ V. Without solving the circuit, use KCL at each node to decide which answer is right.